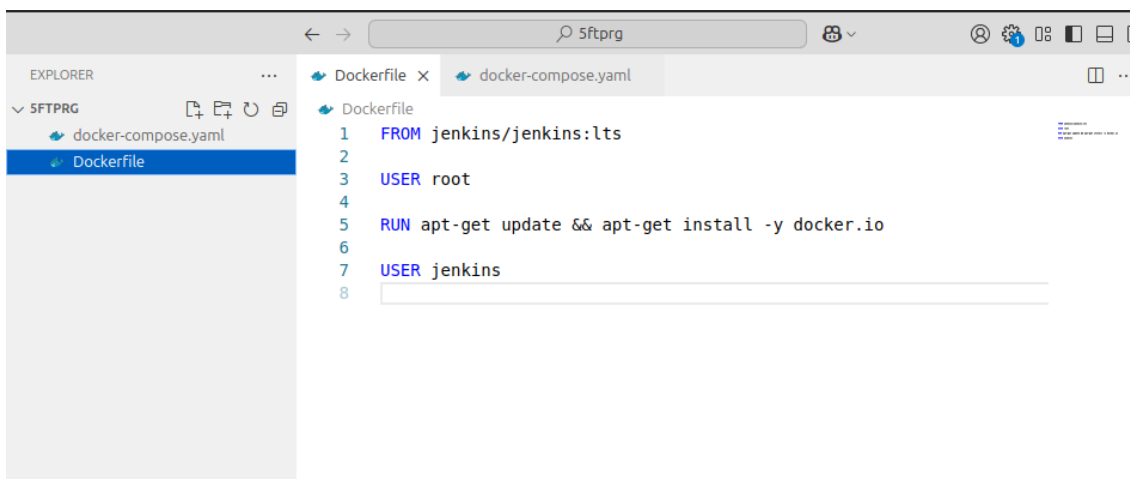


Program 5

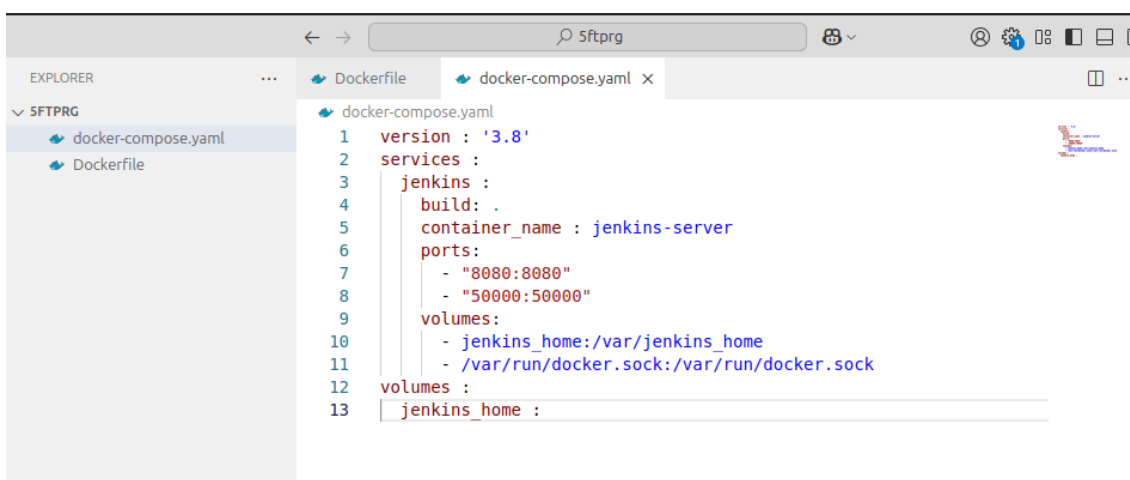
STEP 1: Create Project Directory

```
1rv24mc077_prateek_bhandari@prateek:~$ cd dockerprgs/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ mkdir 5ftprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 5
Command 'cd5' not found, but can be installed with:
sudo apt install cd5
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 5ftprg/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ touch Dockerfile
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ touch docker-compose.yaml
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ docker --version
Docker version 29.0.2, build 8108357
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ docker compose version
Docker Compose version v2.40.3
```

STEP 2: Create Dockerfile



STEP 3: Create docker-compose.yaml



STEP 4: Build & Run Jenkins

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ docker compose up --build
WARN[0000] /home/1rv24mc077_prateek_bhandari/dockerprgs/5ftprg/docker-compose.yaml: the attribute `version` is obsolete, it
[+] Building 45.8s (9/9) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 557B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 139B                                      0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts         4.5s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io          0.0s
=> [internal] load .dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2 12.1s
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2b 0.0s
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa81 12.71kB / 12.71kB 0.0s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed6 1.23kB / 1.23kB 2.0s
=> => sha256:2f1ecb09fa362ebe652038ecbfc64a84b042991e9dedb00059a4127c 5.24MB / 5.24MB 1.7s
=> => sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13 3.12kB / 3.12kB 0.0s
=> => sha256:5687d4df0728e358596e7d718163b2e812a00cf5dd2624459c3315 65.08MB / 65.08MB 5.1s
=> => sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdfBe03923e 2.58kB / 2.58kB 0.0s
=> => sha256:e0e73f34dee9332ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31 183B / 183B 2.8s
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35 95.32MB / 95.32MB 10.5s
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0 189B / 189B 3.9s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221a 6.29MB / 6.29MB 5.6s
```

```
jenkins-server | 2026-01-04 05:31:54.743+0000 [id=34] INFO Winston.Logger#logInternal: Winston Service Engine Running: Controller (disabled)
jenkins-server | 2026-01-04 05:31:55.926+0000 [id=33] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.3
jenkins-server | 2026-01-04 05:31:55.062+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
jenkins-server | 2026-01-04 05:31:55.086+0000 [id=41] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
jenkins-server | 2026-01-04 05:31:56.223+0000 [id=40] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
jenkins-server | 2026-01-04 05:31:56.230+0000 [id=40] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
jenkins-server | 2026-01-04 05:31:56.232+0000 [id=46] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
jenkins-server | 2026-01-04 05:31:56.484+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
jenkins-server | 2026-01-04 05:31:56.485+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
jenkins-server | 2026-01-04 05:31:56.485+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
jenkins-server | 2026-01-04 05:31:56.489+0000 [id=45] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
jenkins-server | 2026-01-04 05:31:56.543+0000 [id=61] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
jenkins-server | 2026-01-04 05:31:57.138+0000 [id=40] INFO jenkins.install.SetupWizard#init:
jenkins-server | [LF]>
jenkins-server | [LF]> *****
jenkins-server | [LF]> *****
jenkins-server | [LF]> *****
jenkins-server | [LF]>
jenkins-server | [LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
jenkins-server | [LF]> Please use the following password to proceed to installation:
jenkins-server | [LF]>
jenkins-server | [LF]> 1257503d5c3c44dfb89cee2686fdd941
jenkins-server | [LF]>
jenkins-server | [LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
jenkins-server | [LF]>
jenkins-server | [LF]> *****
jenkins-server | [LF]> *****
jenkins-server | [LF]> *****
jenkins-server | [LF]>
jenkins-server | 2026-01-04 05:32:03.918+0000 [id=40] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
jenkins-server | 2026-01-04 05:32:03.958+0000 [id=33] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
jenkins-server | 2026-01-04 05:32:06.265+0000 [id=61] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenJ
jenkins-server | 2026-01-04 05:32:06.265+0000 [id=61] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
```

STEP 5: Confirm Jenkins is Running

STEP 6: Get Jenkins Admin Password

STEP 7: Verify Docker Inside Jenkins

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
7706d05f7d79   5ftprg-jenkins "/usr/bin/tini -- /u..." 2 minutes ago   Up 2 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
kns-server    mongo:latest   "docker-entrypoint.s..." 8 weeks ago     Up About an hour   0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
rning_mongodb

1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ docker exec -it jenkins-server cat /var/jenkins_home/secrets/initialAdminPassword
1257503d5c3c44dfb89cee2686fdd941

1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ docker exec -it jenkins-server docker --version
Docker version 26.1.5+dfsg1, build a72d7cd

1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/5ftprg$ █
```

STEP 8: Access Jenkins UI

